Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/06/2004
PI Name:	DeLuca, Dominick Ph.D.		
Project Title:	Effect of Microgravity on the Development of the Immune System		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85724	**Congressional District:**	7
Comments:	PI is deceased as of 10/26/2007 (info from A. Chu/ARC; 9/16/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	08/01/1999	**End Date:**	07/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Witten, Mark (University of Arizona)		
Grant/Contract No.:	NAG2-1369		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have developed a microgravity organ culture dish system (MOCDS) that allows for the normal development of T cells from hematopoietic stem cell (HSC) progenitors. This system mimics the normal development of T cells outside of the body, away from stress hormones made during flight. The MOCDS can be used both on the clinostat and during spaceflight. The culture systems are easy to set up and manipulate, and they should be readily adaptable to the Group Activation Pack (GAP) for use in the Commercial Generic Bioprocessing Apparatus - Isothermal Containment Module (GBA-ICM) Payload. Our work has revolved around studying the effects of vector-averaged gravity, as produced in the clinostat, on the development of T cells in organ culture. In our previous work utilizing the MOCDS we demonstrated the gravity-dependence of T cell development. The results from this work suggested that the full sequelae of pre-TCR signaling may not be taking place in vector-averaged gravity, as indicated by a block at the immature single positive stage of development. Since then, we have set out to determine what the underlying mechanism was for this block.		

	Additionally, using the information obtained from the mechanistic studies, we aimed to reverse the effects that we observed after exposure to clinorotation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work has shed light on the signal dampening effects of microgravity exposure. By further understanding the underlying mechanism of this microgravity-induced effect, we could initiate a new line of research centered on the treatment and prevention of autoimmune disorders, many of which involve signaling defects. For instance, microgravity exposure led to a significant increase in the production of TNF-alpha. Other work in our laboratory is aimed at investigating the role of TNF-alpha during T cell development in the nonobese diabetic (NOD) mouse model, and how over expression of this cytokine may lead to deficient negative selection and hence the production of potentially autoreactive T cells.
Task Progress:	In our current work, we were able to show that after 4 days of vector-averaged gravity exposure fetal thymus organ cultures (FTOC) exhibited a significant reduction in CD4+CD8+ double positive (DP) cell production. Furthermore, CD4-CD8- double negative (DN) cells expressed lower levels of CD3 on their surface. DN, as well as immature single positive (ISP) cells, also expressed reduced levels of the IL-7 receptor alpha chain (CD127). These reductions in CD3 and CD127 expression were concomitantly associated with an increased production of tumor necrosis factor (TNF)-alpha. We were also able to show that addition of an exogenous signal (anti-CD3epsilon monoclonal antibody) to these cultures effectively mitigated the modeled microgravity (MMG)-induced effects, therefore suggesting that MMG-exposure, does in fact, cause a signal dampening effect on developing thymocytes.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Articles in Peer-reviewed Journals	Woods CC, Banks KE, Lebsack TW, White TC, Anderson GA, Maccallum T, Gruener R, DeLuca D. "Use of a microgravity organ culture dish system to demonstrate the signal dampening effects of modeled microgravity during T cell development." Dev Comp Immunol. 2005;29(6):565-82. Epub 2005 Jan 24. PMID: 15752552 , Mar-2005
Dissertations and Theses	Woods CC. "The effects of actual microgravity and vector-averaged gravity on the development of t cells." Dissertation, University of Arizona, March 2004. , Mar-2004